

Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course: Artificial Intelligence

Program: BS(Computer Science)

Duration: 20 Minutes

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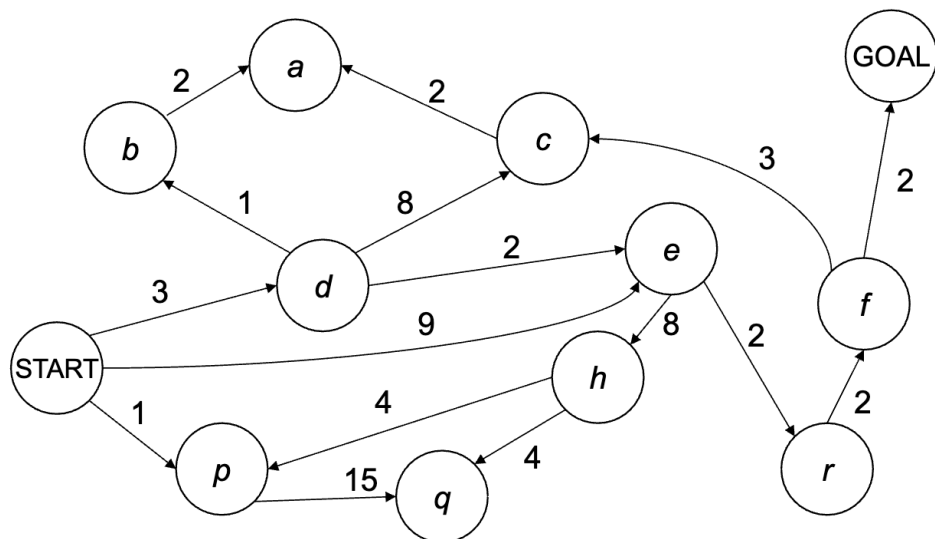
Section: BCS-6C

Total Marks: 10

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Exam: Quiz 1

Q1) Consider the following state space, where each node represents a state, and each directed link represents an action. **(4+6)**



- a) Suppose the state space is to be searched, from "Start" state to "Goal" state. List the nodes of the state space in the order that they are expanded (open list) and visited (closed list) using uniform cost search. **(6)**

	Open	Closed
1	[Start(0)]	[]
2	[p(1), e(9), d(3)]	[Start(0)]
3	[e(9), d(3), q(15)]	[p(1), Start(0)]
4	[e(5), q(15), b(4), c(11)]	[d(3), p(1), Start(0)]
5	[e(5), q(15), c(11), a(6)]	[b(4), d(3), p(1), Start(0)]
6	[q(15), c(11), a(6), r(7)]	[e(5), b(4), d(3), p(1), Start(0)]
7	[q(15), c(11), r(7)]	[a(6), e(5), b(4), d(3), p(1), Start(0)]
8	[q(15), c(11), f(9)]	[r(7), a(6), e(5), b(4), d(3), p(1), Start(0)]
9	[q(15), c(11), Goal(11)]	[f(9), r(7), a(6), e(5), b(4), d(3), p(1), Start(0)]
10	[q(15), Goal(11)]	[c(11), f(9), r(7), a(6), e(5), b(4), d(3), p(1), Start(0)]

b) Draw a search tree from the given state space.

(4)

